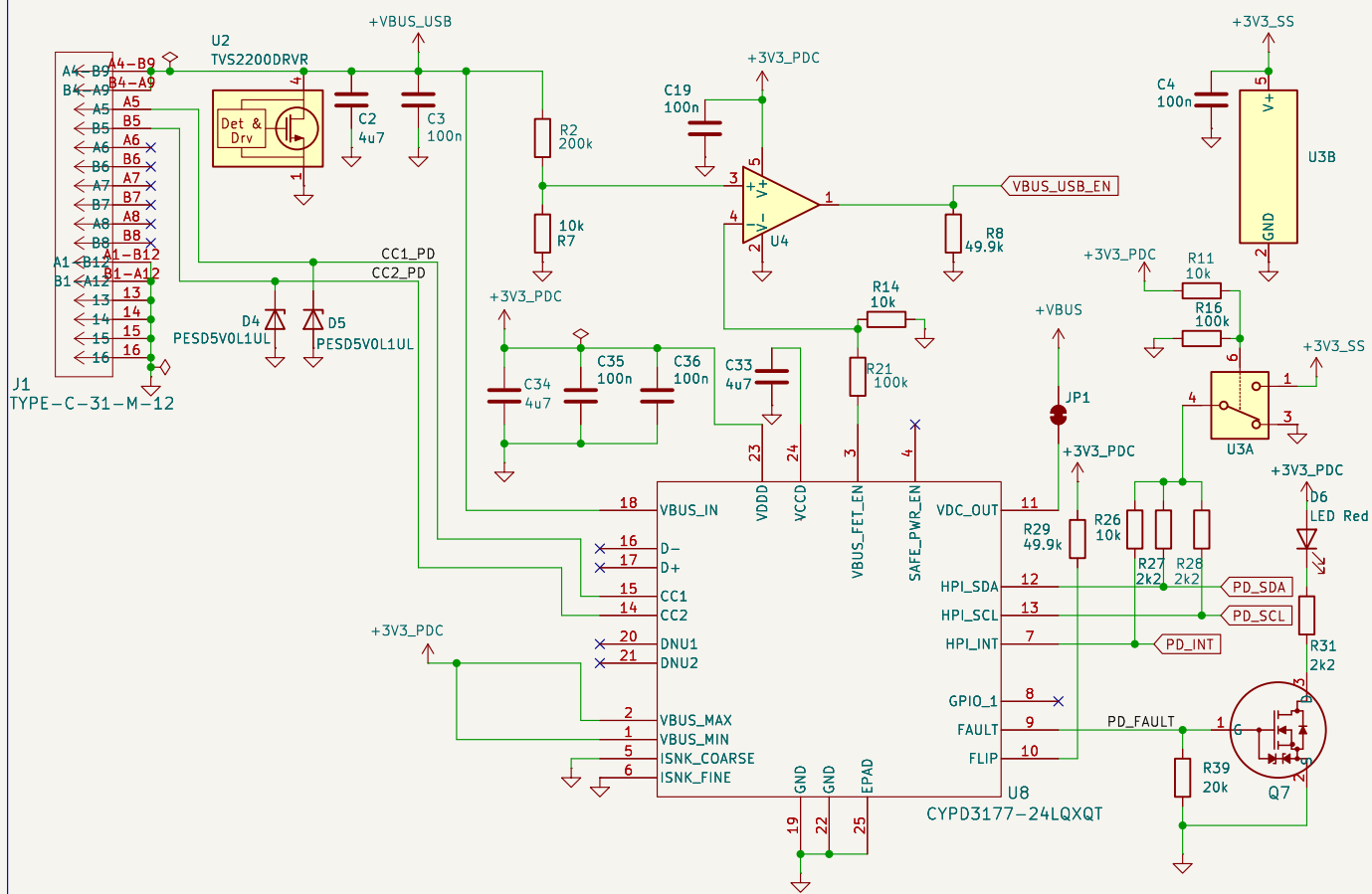
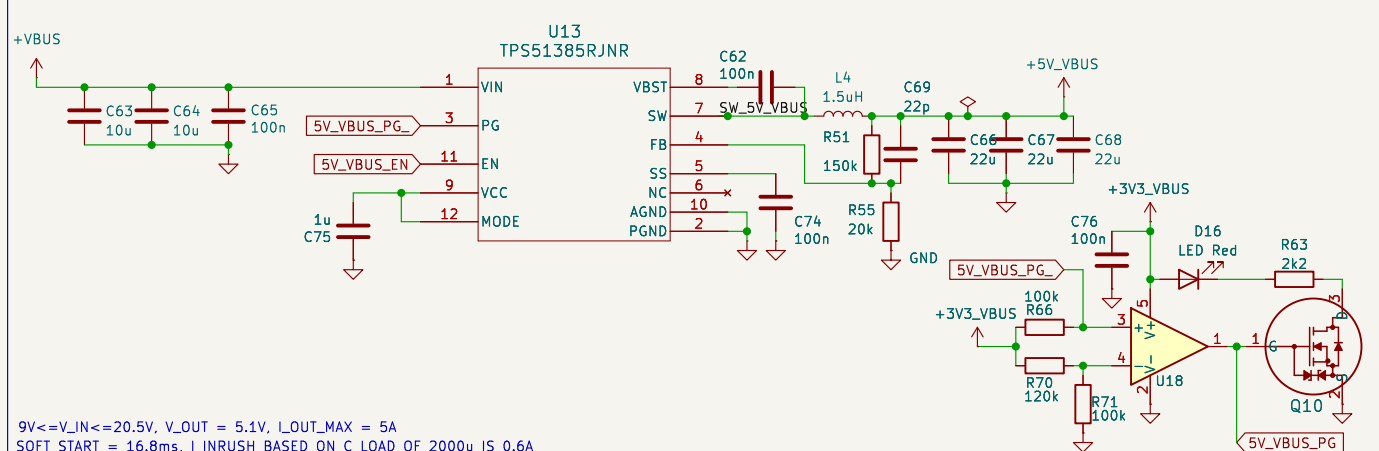


USBC PD



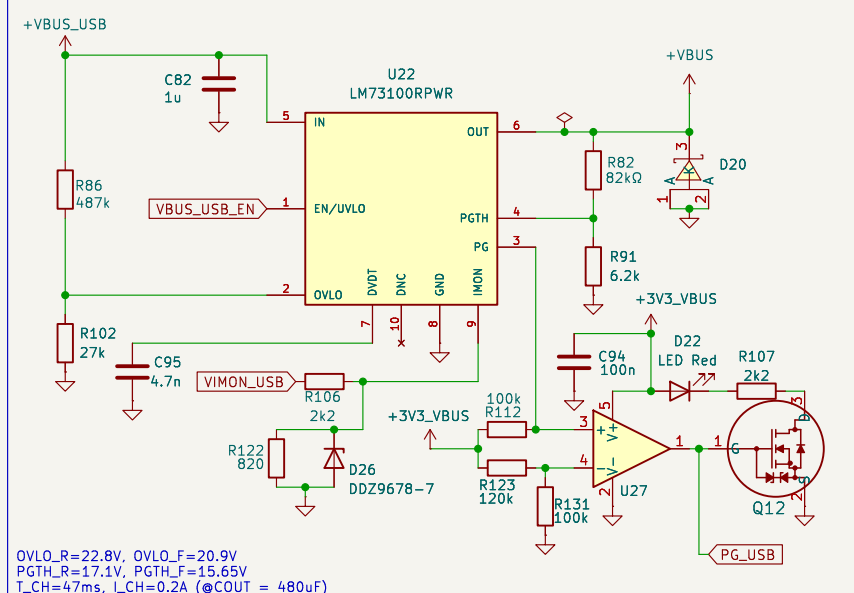
VBUS_MIN,VBUS_MAX = 3V3 -> 20V FROM USBC
ISINK IS NOT GUARANTEED, ANY ADVISED CURRENT IS ACCEPTED, READ BY ESP32 TO VALIDATE

20V->5V BUCK



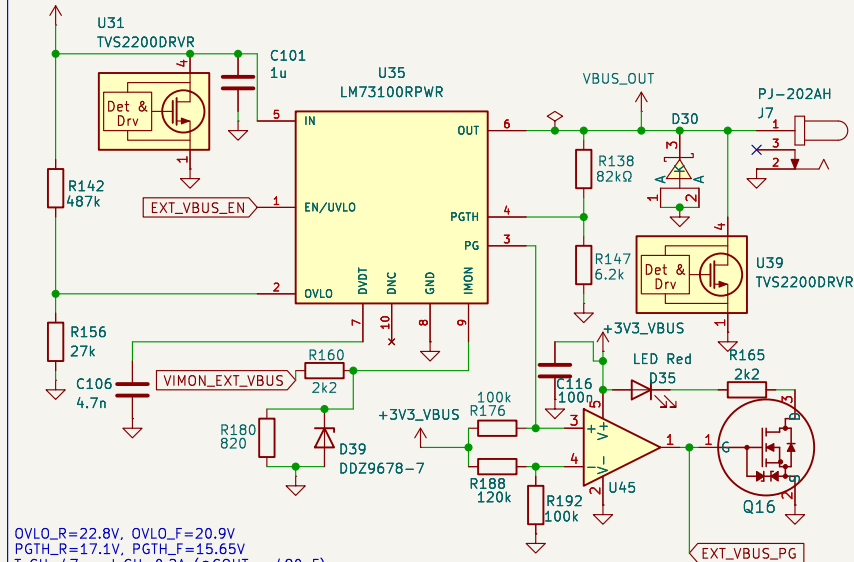
9V<=V_IN<=20.5V, V_OUT = 5.1V, I_OUT_MAX = 5A
SOFT START = 16.8ms, I_LINRUSH BASED ON C_LOAD OF 2000u IS 0.6A

USBC INRUSH CTRL



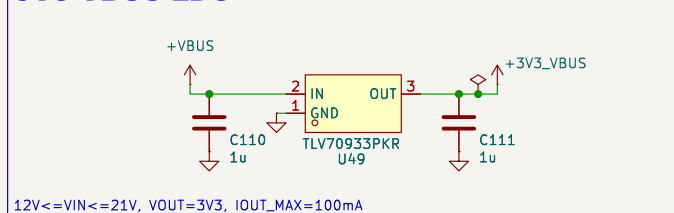
OVLO_R=22.8V, OVLO_F=20.9V
PGTH_R=17.1V, PGTH_F=15.65V
T_CH=47ms, L_CH=0.2A (@COUT = 480uF)

VBUS POWER SWITCH



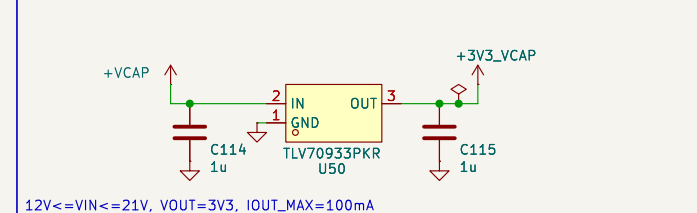
OVLO_R=22.8V, OVLO_F=20.9V
PGTH_R=17.1V, PGTH_F=15.65V
T_CH=47ms, L_CH=0.2A (@COUT = 480uF)

3V3 VBUS LDO



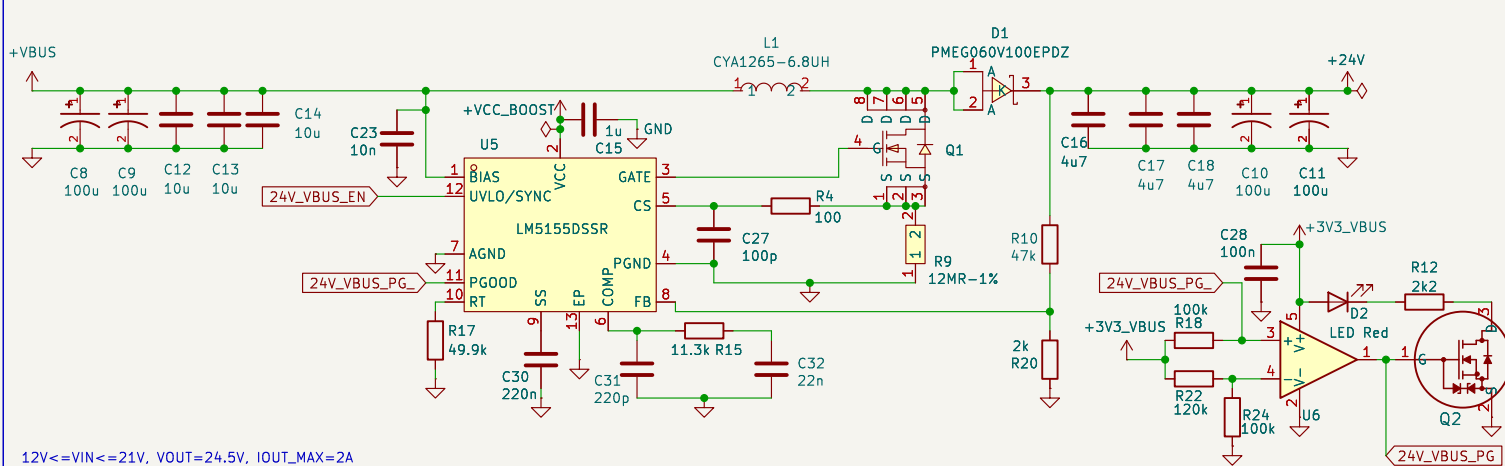
12V<=VIN<=21V, VOUT=3V3, IOUT_MAX=100mA

3V3 VCAP LDO



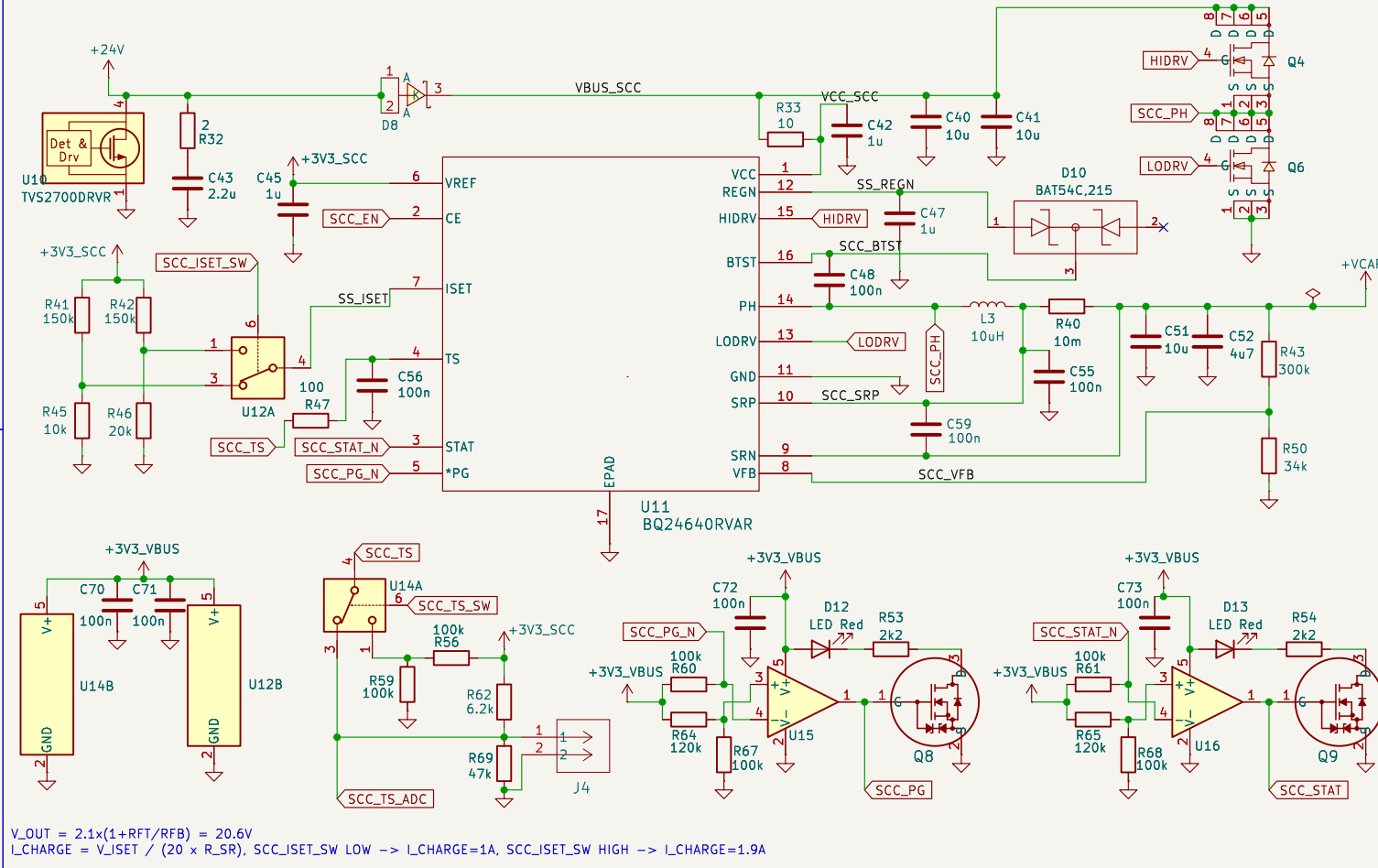
12V<=VIN<=21V, VOUT=3V3, IOUT_MAX=100mA

24.5V BOOST



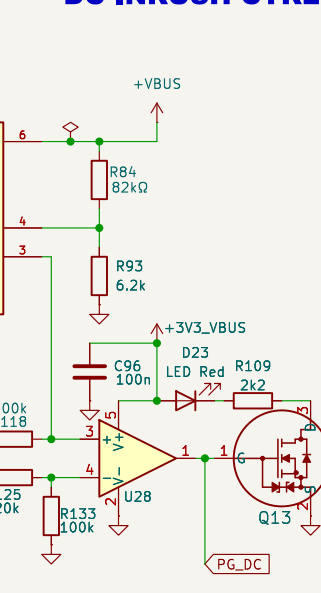
12V<=VIN<=21V, VOUT=24.5V, IOUT_MAX=2A

SUPERCAP CHARGER



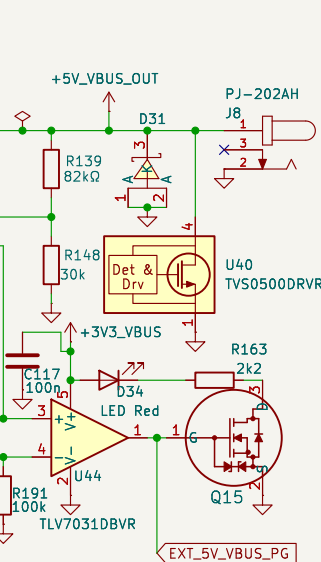
V_OUT = 2.1*(1+R1/R2) = 20.6V
L_CHARGE = V_ISET / (20 * R_SR) = 1A
SCC_ISET_SW LOW -> L_CHARGE=1A, SCC_ISET_SW HIGH -> L_CHARGE=1.9A

DC INRUSH CTRL



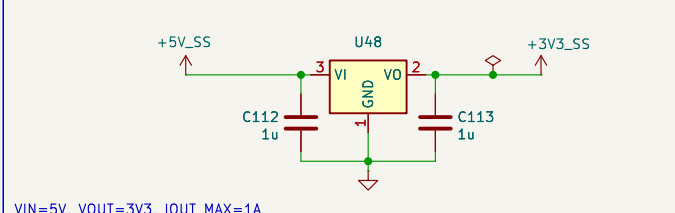
OVLO_R=22.8V, OVLO_F=20.9V
PGTH_R=17.1V, PGTH_F=15.65V
T_CH=47ms, L_CH=0.2A (@COUT = 480uF)

5V VBUS POWER SWITCH



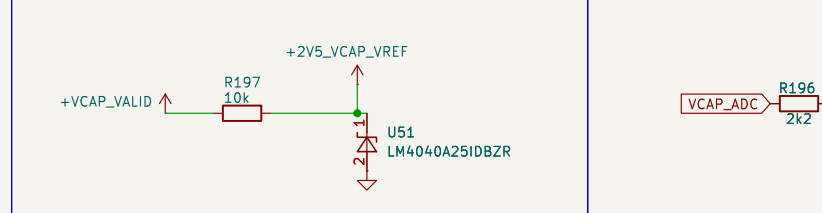
OVLO_R=22.8V, OVLO_F=20.9V
PGTH_R=17.1V, PGTH_F=15.65V
T_CH=47ms, L_CH=0.2A (@COUT = 480uF)

3V3 SS LDO



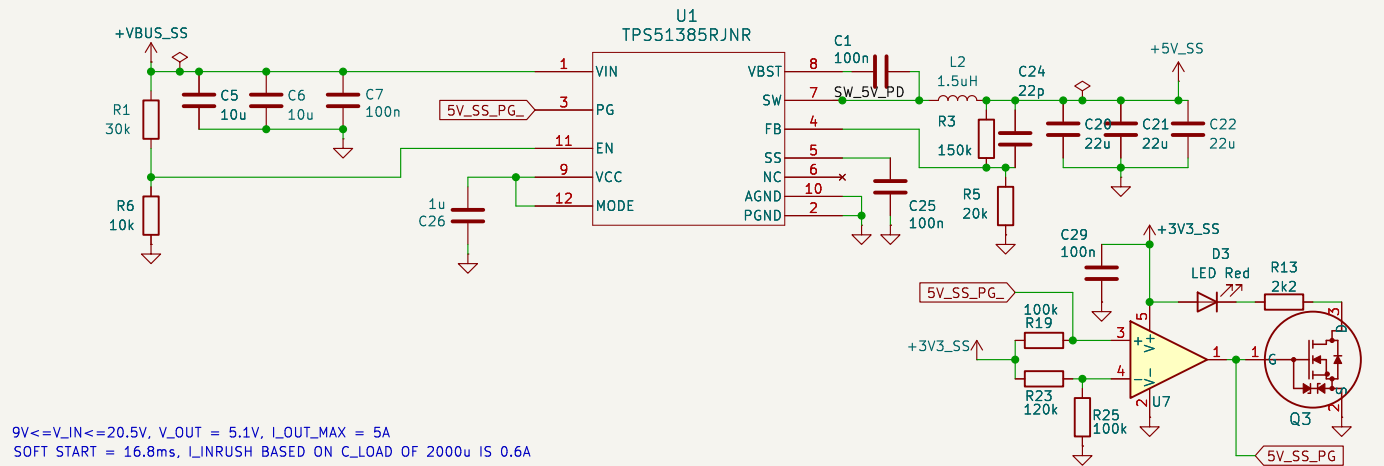
VIN=5V, VOUT=3V3, IOUT_MAX=1A

2V5 VCAP VREF



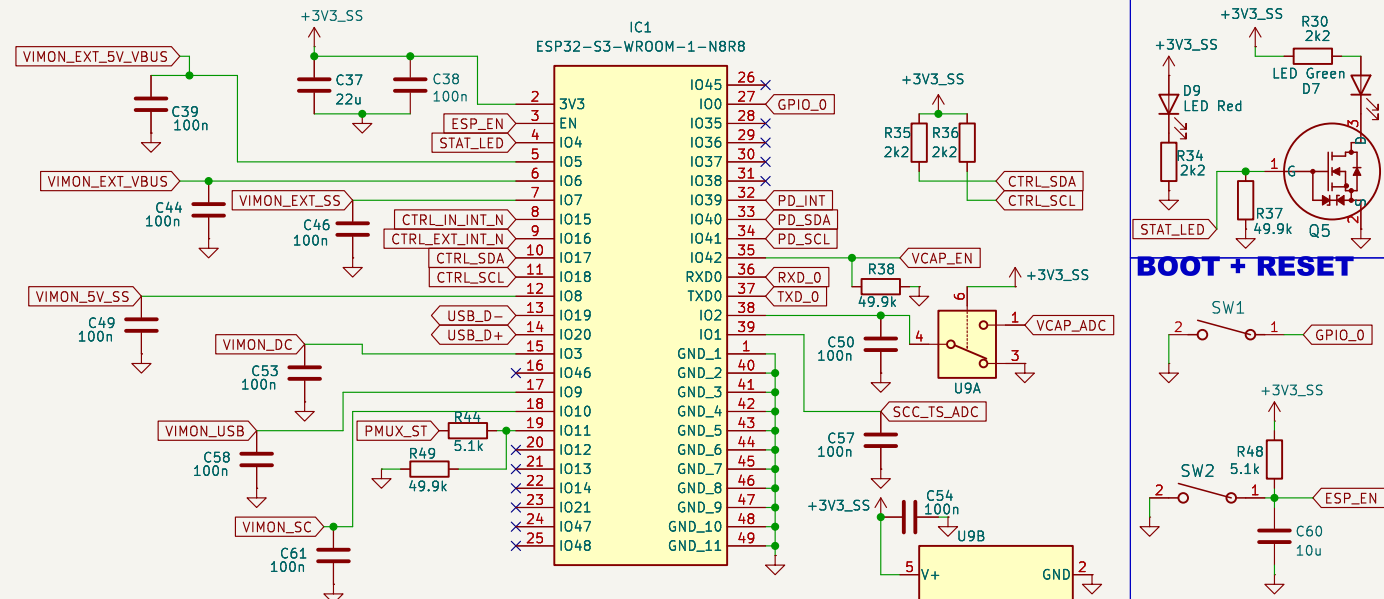
+VCAP_VALID

20V->5V SS BUCK

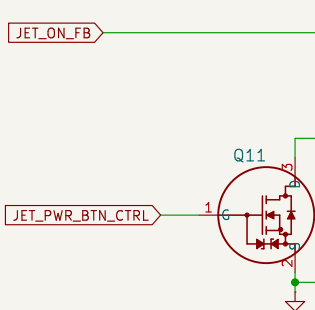


9V<=V_IN<=20.5V, V_OUT = 5.1V, I_OUT_MAX = 5A
SOFT START = 16.8ms, I_LINRUSH BASED ON C_LOAD OF 2000u IS 0.6A

ESP32-S3

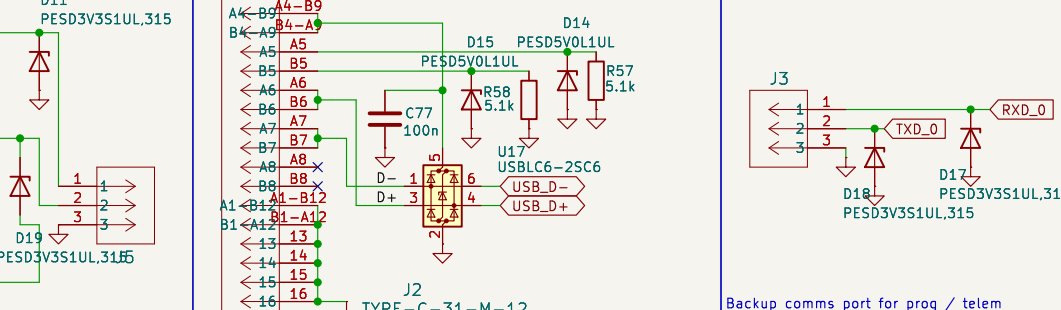


JETSON POWER BTN



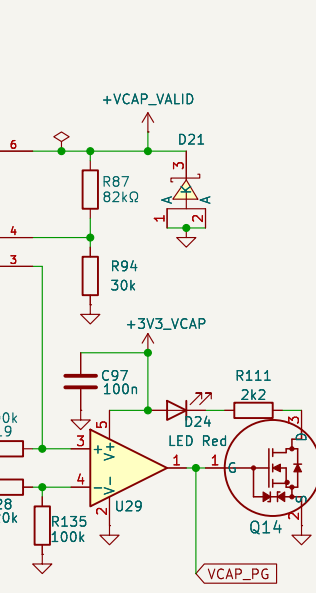
JET_ON_FB

USB-C->ESP32 (NO PWR) ESP32 UART0



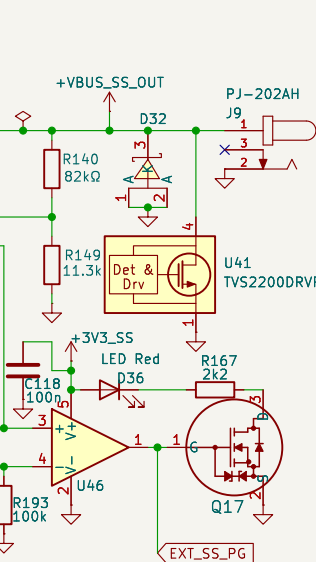
Backup comms port for prog / telem
Primary port is USB-C

SUPER CAP SWITCH



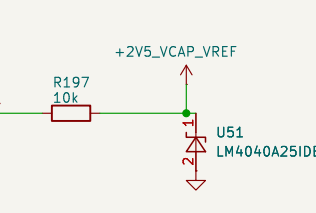
OVLO_R=22.8V, OVLO_F=20.9V
PGTH_R=17.1V, PGTH_F=15.65V
T_CH=47ms, L_CH=0.2A (@COUT = 480uF)

SS POWER SWITCH



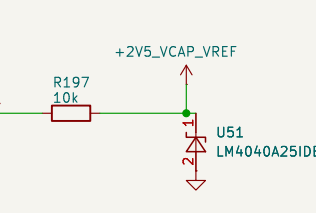
OVLO_R=22.8V, OVLO_F=20.9V
PGTH_R=17.1V, PGTH_F=15.65V
T_CH=47ms, L_CH=0.2A (@COUT = 480uF)

5V SS POWER SWITCH



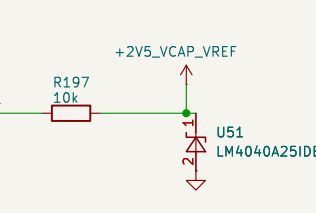
OVLO_R=7.0V, OVLO_F=6.5V
PGTH_R=4.5V, PGTH_F=4.1V
T_CH=84ms, L_CH=0.23A (@COUT = 4000uF)

SS POWER SWITCH



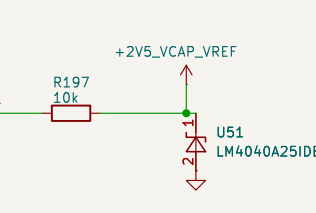
OVLO_R=22.8V, OVLO_F=20.9V
PGTH_R=17.1V, PGTH_F=15.65V
T_CH=47ms, L_CH=0.2A (@COUT = 480uF)

SS POWER SWITCH



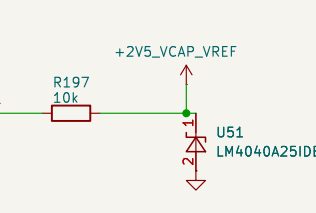
OVLO_R=22.8V, OVLO_F=20.9V
PGTH_R=17.1V, PGTH_F=15.65V
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SS POWER SWITCH



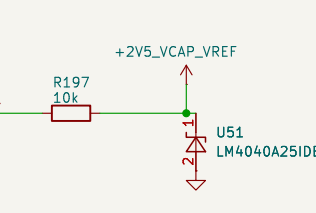
OVLO_R=22.8V, OVLO_F=20.9V
PGTH_R=17.1V, PGTH_F=15.65V
T_CH=47ms, L_CH=0.2A (@COUT = 480uF)

SS POWER SWITCH



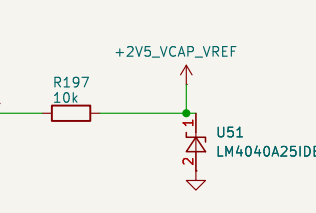
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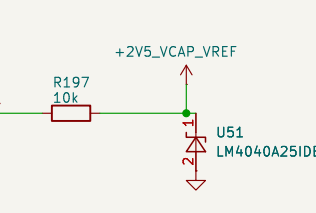
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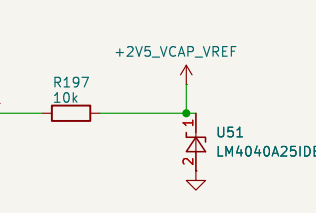
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SS POWER SWITCH



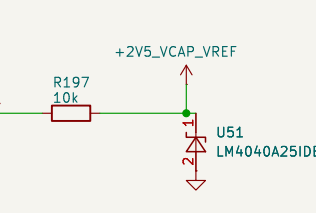
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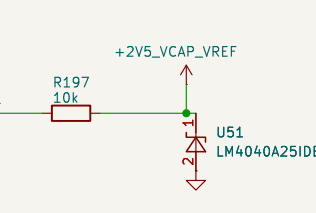
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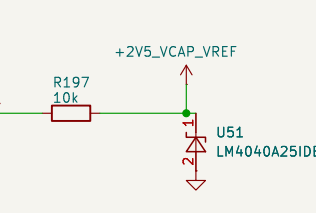
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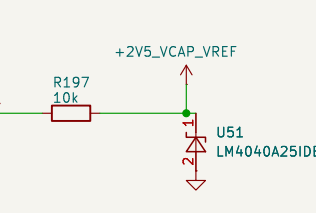
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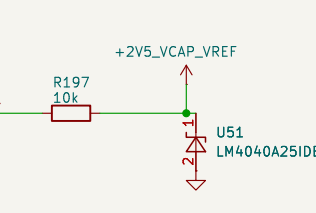
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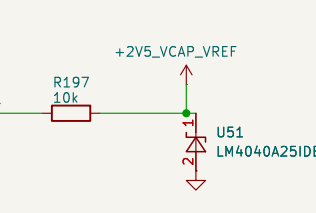
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PGTH_R=17.1V, PGTH_F=15.65V
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SS POWER SWITCH



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PGTH_R=17.1V, PGTH_F=15.65V
T_CH=47ms, L_CH=0.2A (@COUT = 480uF)